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Research

Population genomics of the gametophyte-only fern *Vittaria appalachiana* provides insights into clonal plant evolution

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Summary

- How asexually reproducing organisms maintain genetic diversity and adaptive potential is a long-standing question in evolutionary biology. Asexual lineages have historically been thought of as evolutionary dead ends, yet some exhibit remarkable persistence through time.
- The gametophyte-only fern *Vittaria appalachiana* is a clonal eukaryote, the focus of extensive study due to its peculiar habit and life history, and is an excellent system to explore the consequences of asexuality. Using reduced representation sequencing and life cycle simulations, we assess theoretical expectations for genomic consequences of long-term asexual reproduction and test hypotheses about its origin and demographic history.
- We show that *V. appalachiana* colonies are not patches of single genotypes but are mosaics of genetic diversity, and the accumulation of mutations in the absence of recombination plays an important role in driving this diversity. We identify increased genomic variation, excess heterozygosity, decreased population differentiation, and increased effective population size, all of which are consistent with the expectations for prolonged clonality. Our analyses support the hypothesis that the loss of sexual reproduction in *V. appalachiana* occurred during the Last Glacial Maximum.
- Our results from empirical and simulation-based analyses illuminate how an asexual eukaryote generates, retains, and partitions genomic diversity.

Introduction

All land plants alternate between multicellular diploid (sporophyte) and haploid (gametophyte) life phases. The transition from gametophyte- to sporophyte-dominance is one of the hallmarks of plant evolution (Qiu *et al.*, 2012), and variation in either of these phases can have a profound impact on the evolutionary trajectory of a population (Sorojsrisom *et al.*, 2022). Several modifications of the traditional alternation of generations life cycle exist, including the complete loss of meiosis in total clonal reproduction (hereafter referred to as asexual reproduction). Although asexual eukaryotes are often viewed as evolutionary dead ends (Simon *et al.*, 2003), the loss of one life stage does not necessarily impact the long-term survival of the species (Farrar, 1967); in fact, the transition from sexual to asexual reproduction is not uncommon and has occurred several times across diverse plant lineages (Holsinger, 2000; de Witte & Stöcklin, 2010). However, the genomic features of asexually reproducing species (reviewed by Jaron *et al.*, 2021; see also Pelosi *et al.*, 2023a and references therein) do have profound impacts on population evolution.

In the absence of sexual reproduction, population genetic theory predicts that asexual lineages will experience (1) increased genetic diversity, (2) the preservation or increase of heterozygosity (resulting in an excess of heterozygotes), (3) decreases in population differentiation (F_{ST}), and (4) increases in effective allelic population size (Balloux *et al.*, 2003; Halkett *et al.*, 2005). Without recombination, pairs of homologous chromosomes (and therefore alleles) will diverge as they independently accumulate mutations (the Meselson Effect; Birky, 1996; Mark Welch & Meselson, 2000; Halkett *et al.*, 2005). This can lead to increased observed (H_o) relative to expected heterozygosity under random mating (H_e), resulting in negative values of the fixation index, F_{IS} (the inbreeding coefficient due to nonrandom mating within subpopulations; Hartl & Clark, 1989). As variation within individuals increases, the relative amount of variation between populations decreases, resulting in reduced F_{ST} values. In contrast with sexually reproducing organisms, genetic diversity cannot be lost within clonal individuals through meiotic recombination, resulting in increased effective allelic population sizes (Balloux *et al.*, 2003). While allelic diversity is proportional to the fraction of clonal reproduction, genotypic diversity (i.e. the number of multilocus genotypes) is expected to decrease (Balloux *et al.*, 2003), asexuals can therefore harbor increased diversity of alleles at a locus while having fewer genotypes overall. Empirical data largely support these predictions (Koffi *et al.*, 2009; Ho

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et al., 2019; Reynes *et al.*, 2021), although most existing literature focuses on facultative asexuals with mixed reproductive strategies.

The classic alternation of generations plant life cycle is well exemplified in ferns, which have nutritionally and ecologically independent sporophyte and gametophyte life phases (Haufler *et al.*, 2016). Fern gametophytes have been understudied relative to sporophytes, partly due to their small size and ephemeral nature in temperate regions, but the gametophyte stage plays a crucial role in the evolutionary trajectory of plants (Sorojsrisom *et al.*, 2022) and, as the sexual phase of the life cycle, is critical for establishing new sporophyte populations. Understanding the population dynamics of this crucial portion of the life cycle is essential for a more comprehensive assessment of population genetics and evolution across the plant clade. The breakdown of the alternation of generations can lead to the separation of life phases and short-term (Chen *et al.*, 2013a,b; Ebihara *et al.*, 2013) or long-term (Farrar, 1967; Kuo *et al.*, 2017; Pinson *et al.*, 2017) independent evolution of the gametophyte. Such entirely clonal reproduction in ferns, where the haploid gametophyte persists without the sporophyte, is relatively uncommon but has occurred in at least 25 fern taxa and is termed the 'separation of generations' (Pinson *et al.*, 2017). These ferns represent excellent potential models for investigating the genetic and genomic effects of prolonged asexual reproduction in the gametophytic life stage, and the breakdown of the transition between gametophyte and sporophyte phases in plants.

The most notable and well-studied fern exhibiting the 'separation of generations' life cycle is the Appalachian gametophyte, *Vittaria appalachiana* Farrar and Mickel (Fig. 1a,b). This fern persists exclusively in the gametophyte life stage and reproduces through vegetative propagules called gemmae (Farrar & Mickel, 1991). To date, there are only two reported observations of sporophytes, both abortive and with substantial developmental stunting; neither individual produced spores (Farrar, 1978; Caponetti *et al.*, 1982). Previous researchers have investigated *V. appalachiana*'s ecology (Stevens & Emery, 2015; Chambers & Emery, 2016), physiology (Chambers *et al.*, 2017), phylogenetics (Pinson & Schuettpelz, 2016), population genetics (Farrar, 1985, 1990), and comparative genomics (Grusz *et al.*, 2016; Pelosi *et al.*, 2023a). Farrar (1990) hypothesized that the loss of the sporophyte stage and sexual reproduction was likely related to the cooling climate and glaciation in eastern North America over the last 2 million years (i.e. glaciation events in the Miocene *c.* 2 million years ago (Ma), up to the Last Glacial Maximum *c.* 20 thousand years ago (ka)). *Vittaria* is a tropical genus, and sporophytes of *V. appalachiana* in eastern North America were presumably unable to withstand the onset of freezing temperatures; however, gametophytes of *V. appalachiana* are thought to have persisted in sandstone rockhouses of the Appalachian Mountains, in which temperature and humidity are buffered relative to the environment outside (Farrar, 1998; Stevens & Emery, 2015). Accumulating evidence suggests that *V. appalachiana* likely originated from a polyploidy event (Gastony, 1977; Farrar, 1990; Pinson & Schuettpelz, 2016; Pelosi *et al.*, 2023a,b) although the mode of polyploidization is debated, with results from allozyme studies suggesting an allopolyploid origin (Farrar, 1985, 1990; D. R. Farrar, pers. comm.), while a phylogenetic approach supported an autopolyploid origin (Pinson & Schuettpelz, 2016). Divergence from *Vittaria graminifolia* (*V. appalachiana*'s closest living relative) is estimated to have occurred sometime between 8 Ma (Testo & Sundue, 2016) and 1.4 Ma (Nitta *et al.*, 2022), which accords with Farrar's glaciation hypothesis for a more recent loss of sexuality and the sporophyte. Three outstanding questions remain about the origin of *V. appalachiana* and its asexual habit: (1) the timing of the loss of sexual reproduction, (2) the nature of the polyploid speciation event (i.e. allopolypoidy or autopolyploidy), and (3) whether the forces thought to shape population structure in asexual organisms have acted as expected in this haploid, gametophyte-only fern.

Here, we use ddRADseq and life cycle simulations to explore the population genomics of *V. appalachiana*, assess long-standing questions about its origin and persistence, and interrogate theoretical expectations for population genetics following long-term asexual reproduction. In particular, we assess whether the expected population genetic consequences of prolonged clonal reproduction, including (1) high genomic diversity, (2) excess heterozygosity, (3) decreased population differentiation, and (4) increased effective population size, are observed in this system. We then compare our empirical results to findings from other asexual systems and sexually reproducing ferns. Our study is the first to use fine-scale genome-wide markers to illuminate the impacts of prolonged asexual reproduction in the gametophytic life stage on the population genetics of a fern.

Materials and Methods

Sampling

In this study, we use the following terminology: *Individuals* are plants in one species not connected to other plants in the same species in the same colony; a *colony* is a contiguous or semi-contiguous assemblage of individuals where individuals are separated by no > 5 cm of empty space; and a *population* is an assemblage of colonies found within an 8 km radius (similar to the distance used by Farrar, 1990 and Pinson & Schuettpelz, 2016 to delineate populations; Fig. 1c). This terminology is similar to that used by Farrar (1990). In 2020 and 2021, we collected a total of 120 samples roughly 1 cm^2 (each of which consisted of up to 10 individuals – Fig. 1) from 63 colonies and 13 populations across the range of *V. appalachiana* Farrar and Mickel (Table 1; Fig. 2a; Supporting Information Dataset S1).

DNA extraction and sequencing

Individual gametophytes yielded insufficient DNA quantities for library preparation, and thus, several individual gametophytes from the *c.* 1 cm^2 samples were pooled for extractions (*c.* 5–10 gametophytes per sample). At the time of extractions, we relied on the assumption that each colony likely represented a single clonal genotype based on previous studies (Farrar, 1985, 1990). It is very likely that individuals in these small 1 cm^2 patches are

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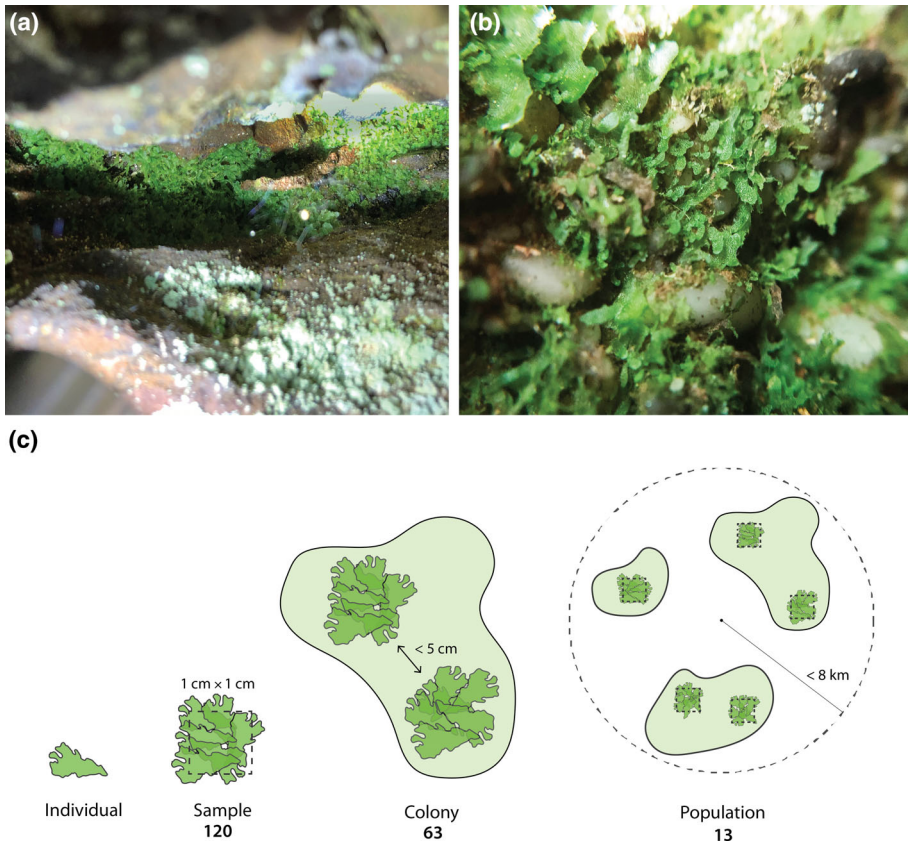


Fig. 1 Sampling of *Vittaria appalachiana*. (a) Colony of *V. appalachiana* from Cloudland Canyon (CC), Georgia. (b) Colony of *V. appalachiana* from Rock City Park (RP), New York at ×10 magnification. (c) Sampling scheme used in this study. Illustration of gametophyte after Farrar & Mickel (1991).

Table 1 Summary statistics for populations of *Vittaria appalachiana*, including the number of samples (*n*), nucleotide diversity (π), rarefied allelic richness (Ar), observed heterozygosity (H_o), expected heterozygosity under Hardy–Weinberg equilibrium (H_S), inbreeding coefficient (F_{IS}), number of private alleles (PA).

Population	*n*	Ar		π		H_o		H_S		F_{IS}		PA	
Daniel Boone (DB)	8	1.136	1.747	0.049	0.149	0.033	0.200	0.028	0.160	−0.175	−0.243	28	144
Pennyrile (PR)	7	1.130	1.703	0.051	0.153	0.035	0.207	0.029	0.161	−0.211	−0.282	44	159
Cloudland Canyon (CC)	8	1.135	1.736	0.052	0.153	0.036	0.210	0.030	0.165	−0.210	−0.267	20	166
Panama Rocks (PN)	4	1.099	1.556	0.048	0.146	0.030	0.191	0.026	0.154	−0.158	−0.236	18	79
Rock City (RP)	3	1.070	1.418	0.047	0.144	0.025	0.177	0.021	0.146	−0.165	−0.212	5	37
Tallulah Gorge (TG)	5	1.107	1.582	0.050	0.150	0.033	0.199	0.028	0.158	−0.190	−0.258	14	85
Noccalula Falls (NF)	3	1.084	1.474	0.049	0.147	0.031	0.184	0.026	0.152	−0.174	−0.227	12	113
Ramsey Cascade (RC)	10	1.139	1.791	0.046	0.140	0.029	0.189	0.025	0.154	−0.135	−0.208	32	221
Pine Mountain (PM)	7	1.122	1.698	0.046	0.140	0.028	0.184	0.024	0.150	−0.143	−0.222	31	236
Breaks Park (BP)	8	1.123	1.702	0.042	0.128	0.026	0.168	0.023	0.142	−0.111	−0.181	14	137
Audra (AA)	5	1.111	1.613	0.048	0.147	0.032	0.192	0.027	0.156	−0.159	−0.226	7	53
McConnells Mill (MM)	2	1.058	1.347	0.045	0.138	0.025	0.163	0.021	0.141	−0.164	−0.158	7	53
Sheick Hollow (SH)	12	1.150	1.829	0.048	0.146	0.033	0.196	0.028	0.157	−0.180	−0.246	40	199

Values in the left columns are estimates from the 'all sites' dataset and those in the right columns are for the variant-only dataset.

clones or nearly clones, but analysis of individuals will be required to assess this assumption. DNA was extracted from 4 to 15 mg of plant tissue for each sample using a modified 3× CTAB extraction (Pelosi *et al.*, 2024), resuspended in 50 µl 1× TE buffer, and quantified using a Qubit 2.0 Fluorometer (Invitrogen) with the dsDNA Broad Range assay. Extracted DNA was sent to the University of Wisconsin-Madison Biotechnology Center (RRID:SCR_017759) for library preparation and sequencing. To determine the optimal enzyme combination for ddRADSeq, five pairs of restriction enzymes were tested on a single sample. The distribution of fragment lengths was visualized on a TapeStation with the High Sensitivity D1000 ScreenTape and compared. From this optimization step, the *NsiI* and *MspI* enzyme combination yielded the best results and were used for digestion of 96 samples from our initial 120 and sequenced on an Illumina NovaSeq6000 (Illumina, San Diego, CA, USA).

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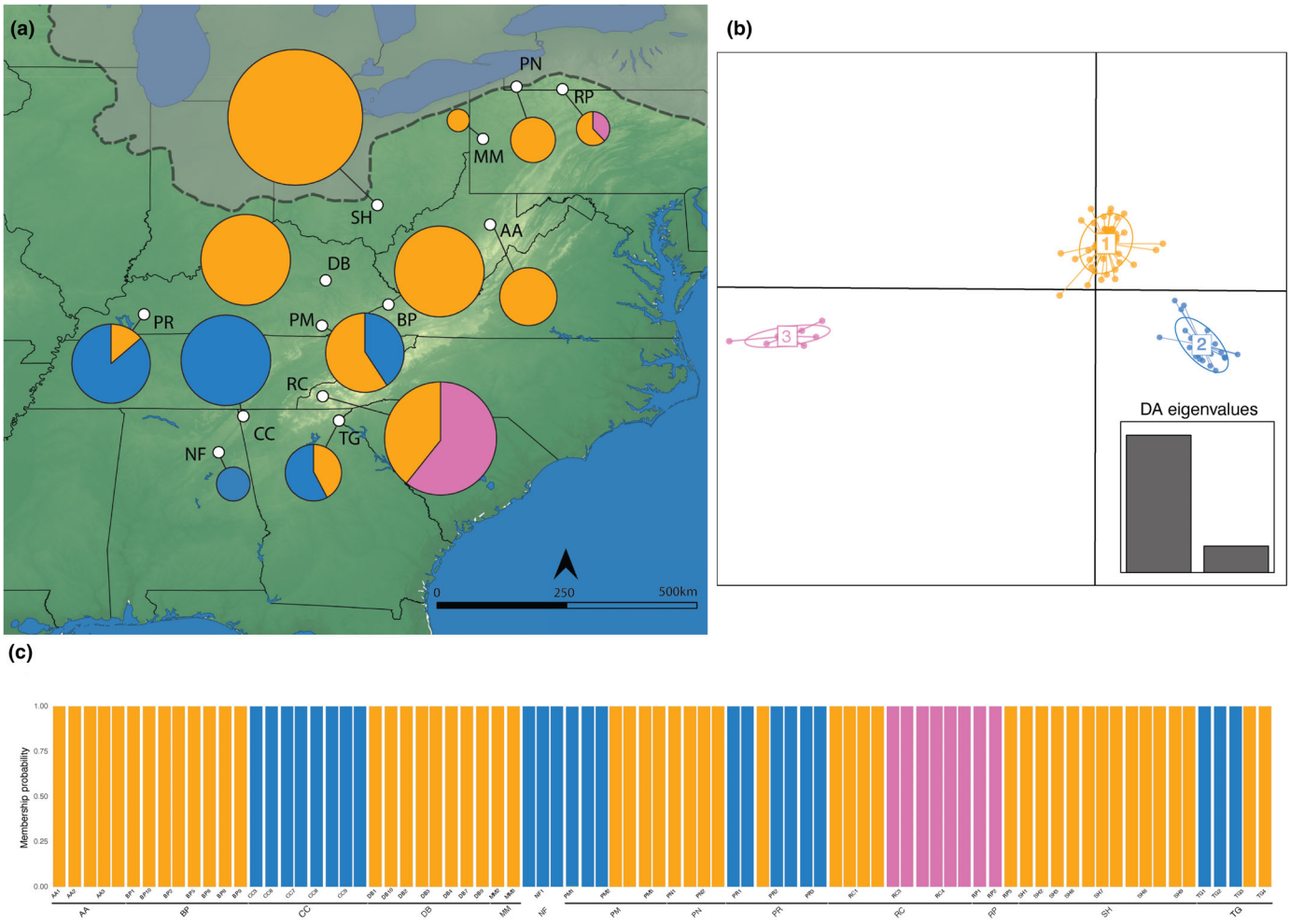


Fig. 2 Genetic clustering of *Vittaria appalachiana*. (a) Map of eastern North America with 13 sampled populations. Population abbreviations are: Daniel Boone (DB), Pennyrile (PR) Cloudland Canyon (CC) Panama Rocks (PN), Rock City (RP) Tallulah Gorge (TG), Noccalula Falls (NF), Ramsey Cascade (RC), Pine Mountain (PM), Breaks Park (BP), Audra (AA), McConnells Mill (MM), and Sheick Hollow (SH). The grayed area with a dashed line indicates the cover and extent of the Laurentide Ice Sheet at the height of the Last Glacial Maximum *c.* 21 thousand years ago (glacial data from Dalton *et al*., 2020). Circle size is proportional to the number of samples; Table 1. (b) Discriminant analysis of principal components (DAPC) of *V. appalachiana* samples, showing three distinct population clusters. (c) Sample assignment to genetic clusters as identified by the DAPC. Colonies are numbered for each population (e.g., DB3 is the third colony sampled in the Daniel Boone population).

Assembly and dataset construction

We used IPYRAD v.0.9.53 (Eaton & Overcast, 2020) to process the raw sequencing data and *de novo* assemble ddRAD loci. We generated an initial assembly with all 96 samples using default parameters. We removed 14 samples represented by < 1000 loci in this initial assembly and re-ran IPYRAD with the remaining 82 samples with a 95% similarity clustering threshold. We then used a workflow starting from the IPYRAD loci to generate a dataset that would allow for the biological peculiarities of *V. appalachiana* and is similar to that used by Kinosian *et al.* (2019). We mapped reads with BWA v.0.7.17 (Li & Durbin, 2009) for each sample to a consensus sequence for each locus generated with EMBOSS v.6.6.0 (Rice *et al.*, 2000). PCR duplicates were marked and removed with GATK v.4.4.0.0 (McKenna *et al.*, 2010), an mpileup file generated and SNPs called with BCFTOOLS v.1.1.9 (Danecek *et al.*, 2021), retaining all variable and invariant sites. We filtered the resulting VCF with VCFTOOLS v.0.1.16 (Danecek *et al.*, 2011) to only keep sites with a minor allele frequency > 0.01, < 50% missing data, and minimum mean read depth of 4. PIXY v.1.2.10.beta2 (Korunes & Samuk, 2021) was used to estimate values of π, F_{ST}, and D_{XY} with invariant sites incorporated into the dataset with a window size of 500 bp. A variant-only dataset was constructed by retaining sites with two or more alleles: *V. appalachiana* has diploid tissue ($n = 120$) relative to the base chromosome number of the genus ($x = 60$) but with respect to Pteridaceae ($x = 30$), there are four homoeologous chromosome copies (or ohnologs). We thinned the datasets by retaining only one SNP for every 500 bp (--thin 500 in VCFTOOLS, approximately one SNP per locus) for each dataset to reduce linkage between sites. An analysis based on the variants from the initial IPYRAD output is presented in Notes S1 and Table S1.

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Plastid population genetic analysis

We used RADOrgMiner v.0.9 (Laczkó *et al.*, 2022) to recover plastid loci from the demultiplexed and trimmed reads. Briefly, RADOrgMiner uses Bwa v.0.7.17 (Li & Durbin, 2009) to map reads to a reference plastome and then calls haplotypes using Freebayes v.1.3.2 (Garrison & Marth, 2012). The *V. appalachiana* plastome (Robison *et al.*, 2018, NCBI accession NC_040219.1) was used as the reference and we used a minimum locus coverage of 15× for a locus to be considered. A second iteration of RADOrgMiner was run with samples with > 10 000 reads mapped to the plastome to avoid spurious haplotype calls. Haplotype networks were constructed with pegas v.1.1 (Paradis, 2010) and haplotypes v.1.1.2 (Aktas, 2020), and population genetic statistics were calculated with mmod v.1.3.3 (Winter, 2012). We replaced negative ϕ_{ST} values for loci with 0, as these are computational artifacts and not biologically meaningful. All analyses were done in R v.4.1.3 (R Core Team, 2022).

Empirical estimates of genomic diversity and differentiation

Estimates of observed and expected heterozygosity and Wright's *F*-statistics were calculated in hierfstat v.0.5-11 (Goudet, 2005) using populations as the main division. Rarefied allelic richness was calculated with pegas v.1.2 (Paradis, 2010) using the extrapolation method as described by Foulley & Ollivier (2006). We used poppr v.2.9.3 (Kamvar *et al.*, 2015) to analyze the hierarchical partitioning of genetic variation with AMOVA (Excoffier *et al.*, 1992). Significance values of the components of covariance were determined using 999 permutations as described by Excoffier *et al.* (1992). Given the high levels of genetic diversity within colonies, correction for clones was not appropriate (but see Notes S2; Fig. S1 for an analysis on clone-corrected data).

We also examined within-colony genetic diversity by generating SNP datasets from the nuclear datasets for six contiguous colonies from different populations (PN2, CC9, RC1, PM2, AA3, and SH7) with at least three samples using Vcftools v.0.1.16 (Danecek *et al.*, 2011). We estimated windowed nucleotide diversity metrics (window size = 500 bp) for each colony for both the variant and invariant and variant-only datasets with Pixy v.1.2.10.beta2 (Korunes & Samuk, 2021) as above. We thinned the datasets by retaining only one SNP for every 500 bp for each VCF and calculated observed heterozygosity (H_o), expected heterozygosity of subpopulation under Hardy–Weinberg Equilibrium (H_S), and F_{IS} with hierfstat v.0.5-11 (Goudet, 2005). Heterozygosity here refers to homoeologous heterozygosity (*sensu* Chapman *et al.*, 1979), which is variation stored as unlinked, duplicated loci as a result of polyploidy; as there are no true homologous pairs of chromosomes of *V. appalachiana*, all genetic variation occurs between homoeologs in separate subgenomes (for further details, see Notes S3). Hereafter, homoeologous heterozygosity is referred to as heterozygosity. A Mantel test was used to assess whether there was a correlation between physical and genetic distance of samples within colonies.

Simulations of genetic diversity

We altered the default homosporous pteridophyte model provided in shadie v.1.0.2 (Sorojsrisom *et al.*, 2022) to reflect the unique life cycle of *V. appalachiana*. Briefly, our new *Vittaria* model is an alternation of generations model (two SLiM generations = one full life-cycle) in which the gametophyte generation is diploid and undergoes clonal reproduction such that, by default, the two chromosomes never experience recombination. Under default settings, during the 'diploid' generation no diploids are produced because sexual reproduction does not occur. For more details on the model, see Figs S2 and S3. We used our new *Vittaria* model to simulate the accumulation of genetic diversity from a starting homogenous population (homozygous at every locus) for 20 000 generations (40 000 SLiM generations) while varying mutation rate (low = 5.0×10^{-10} per base per generation, medium = 5.0×10^{-9}, high = 5.0×10^{-8}), frequency of sexual reproduction (none, low = 1.0×10^{-6} per individual per generation (i.e. approximately one sexual event every thousand generations), medium = 1.0×10^{-5}, high = 1.0×10^{-4}), and type of selection (neutral, purifying, and positive). The combination of these parameters used for simulations is reported in Table 3 (see later) and Dataset S2. The chromosome design was the same across all simulations: a single gene, 12 280 bp long, flanked by two large (> 9000 bp) neutral regions. The gene contained three 360-bp exons separated by two 5600-bp introns. These lengths were chosen based on the averages reported for the model fern *Ceratopteris richardii* by Marchant *et al.* (2022). Selection was controlled by the presence of neutral (no selection), deleterious (purifying selection), or beneficial (positive selection) mutations. The default exon and intron settings from shadie were used to define the distribution of fitness effects.

We ran 500 replicate simulations of each parameter combination (Table 3; Dataset S2) with a gametophyte population size of 1000. Simulations were run using shadie v.1.0.2 and SLiM v.4.1 without recapitation. During the simulation, we recorded three measures of heterozygosity at 100-generation (200 SLiM generation) intervals: expected heterozygosity (H_e), observed heterozygosity (H_o), and Watterson's theta (θ). The calculations for H_e and θ used SLiM's built-in functions *calcHeterozygosity* and *calcWattersonsTheta*, respectively. SLiM uses the mutation frequencies in the entire population to calculate theoretical H_e. To calculate H_o, we randomly sampled 10 individuals from the gametophyte population and used SLiM's built-in *calcPairHeterozygosity* to calculate the heterozygosity between the two chromosomes of each individual, and then recorded the mean H_o of the 10 samples. Additionally, we used H_e and H_o to calculate F_{IS} ($1 - H_o/H_e$) and θ to calculate theoretical effective population size as above at each time point.

Reconstructions of demographic history

Genetic clustering of samples was performed with a discriminant analysis of principal components (DAPC) in adegenet v.2.1.7 (Jombart, 2008) for $K = 1$–13. We evaluated the optimal number of clusters based on the Bayesian information criterion (BIC) for each value of *K*. We used the filtered VCF from ipyrad to run

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Table 2 Summary statistics for genetic diversity within colonies of *Vittaria appalachiana*, including the number of samples (n), nucleotide diversity (π), observed heterozygosity (H_o), expected heterozygosity under Hardy–Weinberg Equilibrium (H_S), and inbreeding coefficient (F_{IS}). Colonies are numbered for each population (e.g., AA3 is the third colony sampled in the Audra population).

Colony	n	π		H_o		H_S		F_{IS}	
AA3	3	0.047	0.144	0.029	0.181	0.026	0.150	−0.136	−0.209
CC9	3	0.052	0.157	0.035	0.204	0.029	0.163	−0.208	−0.252
PM2	4	0.047	0.142	0.028	0.185	0.024	0.150	−0.158	−0.233
PN2	3	0.048	0.147	0.029	0.186	0.025	0.152	−0.144	−0.226
RC1	4	0.038	0.117	0.020	0.147	0.019	0.127	−0.084	−0.158
SH7	3	0.054	0.161	0.037	0.213	0.031	0.166	−0.209	−0.281

Values in the left columns are estimates from the 'all sites' dataset and those in the right columns are for the variant-only dataset.

Table 3 Parameter settings and results from SHADIE simulations.

Model	Sexual repr. rate	Mutation rate (μ)	Selection	H_e[a]	H_o[a]	F_{IS}[a]	θ[a]	N_e[a]
Vittaria	None	5E-9	Neutral	5.39E-05	9.63E-05	−0.788	2.93E-05	2934
Vittaria	None	5E-9	Purifying	3.16E-05	5.65E-05	−0.786	1.83E-05	1833
Vittaria	None	5E-9	Positive	3.58E-05	6.49E-05	−0.812	1.94E-05	1937
Vittaria	1E-6	5E-9	Neutral	5.14E-05	8.92E-05	−0.737	2.90E-05	2902
Vittaria	1E-6	5E-9	Purifying	2.93E-05	5.02E-05	−0.713	1.82E-05	1820
Vittaria	1E-6	5E-9	Positive	3.01E-05	5.20E-05	−0.728	1.86E-05	1858
Vittaria	1E-5	5E-9	Neutral	2.74E-05	2.74E-05	−0.219	2.22E-05	2219
Vittaria	1E-5	5E-9	Purifying	1.23E-05	1.44E-05	−0.168	1.38E-05	1379
Vittaria	1E-5	5E-9	Positive	1.33E-05	1.64E-05	−0.234	1.39E-05	1387
Vittaria	1E-4	5E-9	Neutral	9.95E-06	2.33E-06	0.766	1.59E-05	1593
Vittaria	1E-4	5E-9	Purifying	6.21E-06	1.25E-06	0.799	1.02E-05	1024
Vittaria	1E-4	5E-9	Positive	7.50E-06	3.92E-06	0.478	1.05E-05	1045
Homosporous	–	5E-9	Neutral	1.19E-05	1.19E-05	−0.020	1.17E-05	1167
Homosporous	–	5E-9	Purifying	6.91E-06	6.91E-06	0.044	7.40E-06	740
Homosporous	–	5E-9	Positive	7.00E-06	7.00E-06	−0.011	7.02E-06	702

Estimates of expected heterozygosity (H_e), observed heterozygosity (H_o), Watterson's theta (θ), inbreeding coefficient (F_{IS}), and effective population size (N_e) values are reported for generation 10 000. All simulations using the *Vittaria* model were performed with gametophyte census size of 1000 (2000 chromosomes) and simulations using the sexual homosporous model were performed with a gametophyte census size of 2000 and sporophyte census size of 1000 (2000 chromosomes). In the homosporous fern model, there is no sexual reproduction rate paramter, as sexual reproduction occurs every generation. Full parameter settings reported in Supporting Information Dataset S2.
[a]At generation = 10 000.

these analyses (1664 SNPs), as and our other datasets failed to identify an optimal K less than the number of individuals, which may be an artifact from our filtering method. We used easySFS.py (https://github.com/isaacovercast/easySFS) to downsample our variant-only VCF and generate site frequency spectra (SFS) suitable for the reconstruction of the demographic history. Given that individuals can produce gemmae within a year in culture (Farrar, 1990, though reproductive rates are not known in the field), we used a generation time of 1 yr. We estimated a mutation rate of 1.172×10^{-8} mutations per site per generation based on the average synonymous substitution rate and divergence times calculated by Pelosi *et al.* (2023a). This value is *c.* 2.5× greater than the mutation rate estimate from Barker (2009) of 4.79×10^{-9} mutations per site per generation, which aligns with the accelerated rate of molecular evolution in nuclear coding regions identified by Rothfels & Schuettpelz (2014). We used a combination of generation times and mutation rates to explore how effective population size has changed over time (Fig. S4). To infer changes in the effective population size of *V. appalachiana* from the site frequency spectrum (SFS) over time, we used STAIRWAY PLOT 2 v.2.1.2 (Liu & Fu, 2020), which uses a method that can be applied to low-coverage data types, such as RADSeq.

We further compared several demographic models using δaδi v.2.3.3 (Gutenkunst *et al.*, 2009) from Portik *et al.* (2017) (Fig. S5) and treated all samples as belonging to a single population given the low genetic divergence among populations (Table 1). The model parameters were optimized using the dadi_pipeline v.3.1.7 (Portik *et al.*, 2017) with five rounds of 10 replicates for each model using the optimization method from Nelder & Mead (1965) with custom scripts. We compared the AIC scores of the top runs of each model to determine the best fit to the data. To scale the parameter estimates from δaδi, the ancestral effective population size was calculated from the estimated theta parameter as:

$$N_{Anc} = \theta / 4\mu L$$

where N_{Anc} is the ancestral effective population size, θ is the scaled mutation rate, μ is the per-generation per-site mutation

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rate, and L is the total sequence length. Timing of demographic events in generations was scaled by

$$T = \tau \times 2N_{\text{Anc}}$$

where T is the timing of the event in generations and τ is the parameter time estimate from δaδi.

Results

Assemblies and datasets

A total of 229 million PE reads were generated, with an average of 2.204×10^6 PE reads per sample after demultiplexing and filtering. The initial assembly containing all 96 samples identified 137 551 filtered loci with an average of 9860 loci per sample (range: 0–23 480 loci) and 995 407 SNPs. After removing 14 samples with < 1000 loci in the initial assembly, we used the datasets clustered to 95% similarity and identified 208 884 loci. By mapping reads back to consensus sequences for each of these loci and filtering for missing data and depth, we generated two datasets. The variable and invariant site dataset contained 14 167 loci with 1609 347 total sites; the variant-only dataset contained 13 558 loci and 608 173 sites. After thinning the data to include approximately one site per locus, there was a total of 13 588 sites in the variable + invariant dataset and 14 167 sites in the variant-only dataset. We discuss the results of the variant-only dataset here but see the Dataset S1 for information on the variable and invariable dataset and analyses on the ipyrad variants.

An initial alignment totaling 5212 bp was recovered from the plastome, including 17 loci with an average length of 306 bp per locus containing all 96 samples. On average, there were 105 425 reads mapped to the plastome per sample (range: 70–295 584), with 30 823 bp covered (range: 2579–72 245, *c.* 20.6% of the plastome) at 76.0× depth of coverage (range: 0.05×–252.2×). After removing samples with < 10 000 reads mapping, a second alignment was recovered that contained 80 of the initial 96 samples. In this subset, there were on average 126 164 reads mapped to the plastome per sample (range: 16 304–295 584), with 35 851 bp covered (range: 15 919–72 245, *c.* 23.9% of the plastome) at 90.9× depth of coverage (range: 10.2×–252.2×). The alignment contained 9262 bp across 22 loci with an average length of 421 bp per locus. The number of variable sites ranged from 0 to 145; with the highest number of informative sites in the noncoding region between *rpoB* and *trnD*.

Genetic diversity in the plastid was low with a nucleotide diversity (π) equal to 4.976×10^{-4}. Most samples (61 of 80) were represented by two distinct haplotypes (I, III) that differ by a single step with 12 minor haplotypes differing from major ones or each other by two or fewer steps (Fig. S6). Most populations shared haplotypes, and there was low differentiation among populations with $\phi_{ST} = 0.0603$. This value dropped slightly to $\phi_{ST} = 0.0347$ when comparing colonies rather than populations. Major haplotypes were found across populations with the exception of PM, which only contained samples in haplotypes IX, X, and XI (Fig. S6).

Population genomic metrics

Here, we report the results where we consider populations as the major grouping of samples and the variant-only dataset and within-colony datasets; see Dataset S1 for analyses considering colonies as the major division and results of the variable and invariant dataset. As mutations independently accumulate within individuals without recombination, assessments of long-term asexual lineages should reveal (1) heightened genomic diversity as measured by allelic and nucleotide diversity, (2) excess heterozygosity, and (3) low population differentiation.

Allelic and nucleotide diversity The average global rarefied allelic richness (Ar) was 1.630 alleles/locus (range of 1.347 alleles/locus in McConnells Mill (MM) to 1.829 alleles/locus in Sheick Hollow (SH)). The number of private alleles in populations ranged from 37 in Rock City Park (RP) to 236 in Pine Mountain (PM). Nucleotide diversity was generally high; the average π was 0.145 across 500-bp windows. We also recovered a high level of genetic diversity within colonies, with an average windowed nucleotide diversity of 0.144 (Table 2).

Heterozygosity Overall, global levels of observed heterozygosity in *V. appalachiana* were moderate ($H_o = 0.189$) and were slightly greater than expected heterozygosity ($H_S = 0.154$), resulting in heterozygote excess ($F_{IS} = -0.228$; Table 1). The percentage of loci with negative F_{IS} values was 97.2% on average, ranging from 94.3% for MM to 98.4% for Pennyrile (PR). Within colonies, the observed heterozygosity was lower for the variable and invariant dataset (average: 0.0.29; range: 0.020–0.035) than the variant-only dataset (average: 0.186; range: 0.147–0.213) and all colonies had an excess of heterozygotes, with an average F_{IS} of −0.226.

Population differentiation Genetic differentiation of populations of *V. appalachiana* was low ($F_{ST} = 0.017$, $D_{XY} = 0.142$), with population-level pairwise F_{ST} values ranging from 6.36×10^{-3} to 2.66×10^{-2}. A hierarchical AMOVA recovered that nearly all variation occurred within samples (96.208%, $P = 0.001$), with less variation occurring between samples within a population (2.927%, $P = 0.001$), and between populations (0.863%, $P = 0.010$). When examining data from within colonies, we did not find a significant correlation between genetic and physical distance of samples within colonies (linear mixed-effects model, $t = -0.227$, df = 15, $P = 0.823$).

SHADIE Simulations

We simulated 20 000 generations (40 000 SLiM generations) in total, but report the first 10 000 generations here, reflecting the approximate time since the Last Glacial Maximum. The trends exhibited by the first 10 000 generations continue for the duration of the simulation. All reported generation times are 'true' generations (a full life cycle from gametophyte to gametophyte); SLiM generations will be referenced explicitly. In our SHADIE simulations, H_o, H_e, and θ all scaled linearly

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with mutation rate. Genetic diversity by all three measures was consistently higher in neutral simulations compared with simulations with selection; however, there was very little difference in genetic diversity between simulations with purifying selection vs positive selection (Fig. 4, see later). By simulating data under different models and rates of sexual reproduction, we can examine how sexual reproduction is expected to alter metrics of (1) heterozygosity and (2) effective population size and how these metrics compared with our empirical data.

Heterozygosity Very low frequencies of sexual reproduction resulted in slightly depressed genetic diversity, whereas a moderate frequency caused a large decrease in genetic diversity (Fig. 4a, b). Observed heterozygosity was consistently higher than expected heterozygosity in simulations without sexual reproduction, which is reflected by consistently negative values of F_{IS} in these simulations. However, in the presence of sexual reproduction, this pattern starts to be lost once the frequency of sexual reproduction exceeds 10^{-5}. H_o is lower than H_e when sexual reproduction occurs approximately every 10 generations, and F_{IS} becomes very positive (Fig. 4d).

Effective population size The effects of selection and sexual reproduction on effective population size (N_e) were substantial. In all non-neutral simulations, N_e began well below the true census size (1000 diploid individuals), reflecting the lack of genetic diversity at the beginning of the simulation. In neutral simulations with low or no sexual reproduction, effective population size matched the census size immediately and far exceeded the census size by 10 000 generations by almost 300%. N_e was relatively depressed by the presence of selection and sexual reproduction. In all non-neutral simulations, except for those with a high rate of sexual reproduction, N_e approaches true census size *c.* 500 generations. By contrast, in simulations with a high frequency of sexual reproduction, N_e hovers around the true census size of 1000 for most of the simulation (Fig. 4c).

Demographic history reconstruction

The DAPC identified three genetic clusters ($k = 3$, $\Delta BIC = 0.5998$; Figs 2a, S7). Most populations (8 of 13) had samples that were assigned to a single genetic cluster and nearly all colonies (47 of 50) were composed of samples that were assigned to a single genetic cluster (Fig. 2b,c). The demographic inference from the stairway plot analysis shows a gradual increase in effective population size during/following the Last Glacial Maximum *c.* 26–17 ka (Fig. 3a). There is a gradual decrease since that time, possibly related to the gradual extinction of colonies over time. Of the six single population demographic models tested in δaδi, the three epoch model with inbreeding had the best fit ($\Delta AIC = 1.86$; Fig. 3b). The ancestral effective population size (N_{Anc}) was estimated to be 5071 with an increase in size to $N_B = 68\,448$ at 46 212 generations ago, with a decrease in size $N_{Pres} = 38\,200$ at 45 260 generations ago and inbreeding coefficient of 0.74.

Discussion

How asexually reproducing organisms maintain genetic diversity and adaptive potential has been a long-standing question in evolutionary biology (Maynard Smith, 1978). Here, we used the enigmatic gametophyte-only fern *V. appalachiana* as a study system to explore the consequences of prolonged clonal reproduction on population evolution in a gametophyte-only fern. In particular, we assessed whether the expected consequences of long-term clonality, including (1) heightened genomic diversity, (2) excess heterozygosity, (3) decreased population differentiation, and (4) increased effective population size, are observed in this system. We also addressed long-standing questions about the origin of *V. appalachiana* and its asexual habit by assessing (1) the most likely mode of polyploid speciation, (2) the likely timing of the loss of sexual reproduction, and (3) the dominant forces that have shaped current population structuring.

Population genetic consequences of prolonged asexual reproduction

Asexual eukaryotic taxa are often considered evolutionary dead ends (Maynard Smith, 1978), occupying short terminal branches on the tree of life (Johnson, 2006; Neiman *et al.*, 2009; Schwander & Crespi, 2009). Long-term asexual and parthenogenetic reproduction is well understood in other systems, with much of our knowledge coming from animals (Ricci, 1987; Judson & Normark, 1996; Mark Welch & Meselson, 2000; Schwander *et al.*, 2013; Brandt *et al.*, 2017, 2021). A critical assumption underpinning expectations about population genomics and asexual reproduction is that in the absence of meiotic recombination, each chromosome will accumulate mutations independently of its counterpart and that these are not spread within or between populations through gene flow (the Meselson Effect; Birky, 1996; Mark Welch & Meselson, 2001). Here, we place our findings for *V. appalachiana* in the context of the expectations resulting from this phenomenon as described in the existing literature.

Expectation 1: asexuality should lead to higher levels of genetic diversity In their review of clonal plant populations, Ellstrand & Roose (1987) found that genotypic diversity was generally high with few identical individuals and that most clonal populations consisted of multiple clonal lineages. By contrast, Farrar (1990) found that only four of the 92 sampled colonies of *V. appalachiana* had more than one genotype present and, of the colonies he sampled, 85% were monomorphic for their allozyme banding patterns, but highlighted that genetic variation in the species is similar to that of sexually reproducing taxa. These results were based on a single locus, using a method where only nonsynonymous mutations in specific protein-coding regions can be assessed, and the sample analyzed contained hundreds of individuals. Using higher resolution genetic markers and a finer sampling scheme, we recovered surprisingly high levels of genetic diversity within colonies of *V. appalachiana*, and our results suggest that these colonies are not stands of single genotypes, but

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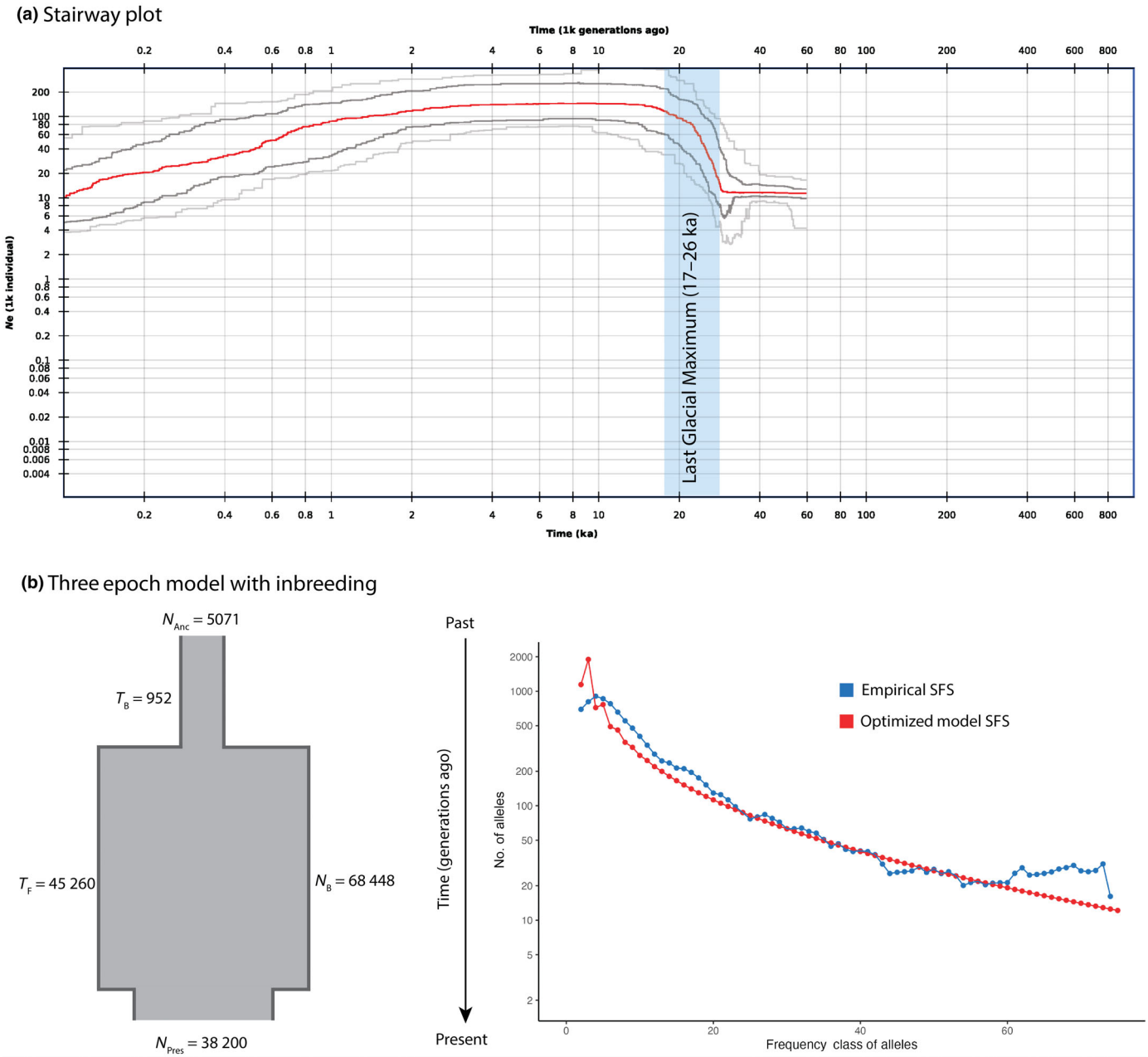


Fig. 3 Demographic history of *Vittaria appalachiana*. (a) Stairway plot of inferred ancestral effective population size of *V. appalachiana* with a mutation rate of 1.172×10^{-8} mutations per site per generation and a generation time of 1 yr. The red line is the median of the inferences, dark gray is the 75% confidence interval, and light gray is the 95% confidence interval. (b) Model parameters inferred using the three epoch with inbreeding model in δaδi and associated empirical (blue) and optimized model site frequency spectra (SFS) (red).

rather are mosaics of genetic diversity. The windowed nucleotide diversity of the within-colony data was $\pi = 0.144$, which is similar to the nucleotide diversity in the sexually reproducing *Athyrium sinense* fern complex (*c.* 0.08; Liang *et al.*, 2022) and greater than that observed in the facultative asexual duckweed *Spirodela polyrhiza* with $\pi = 5.42 \times 10^{-4}$ (Ho *et al.*, 2019), although we note that levels of genetic diversity are likely lineage-dependent. Intra-sample variation also accounted for the vast majority of the variation in the AMOVA, emphasizing the genomic diversity within samples. Similar patterns of high genomic diversity have been documented in several other long-term asexual lineages (García-Cuenca *et al.*, 2016; Bast *et al.*, 2018). We also observed higher values of Watterson's θ, a measure of population genetic diversity, in the *Vittaria* SHADIE simulations with no or low sex compared to the sexual homosporous models, consistent with both our empirical data and theoretical expectation.

Expectation 2: asexual lineages should have excess heterozygosity The accumulation of independent mutations on homologs (or homeologs) should lead to observed increases in

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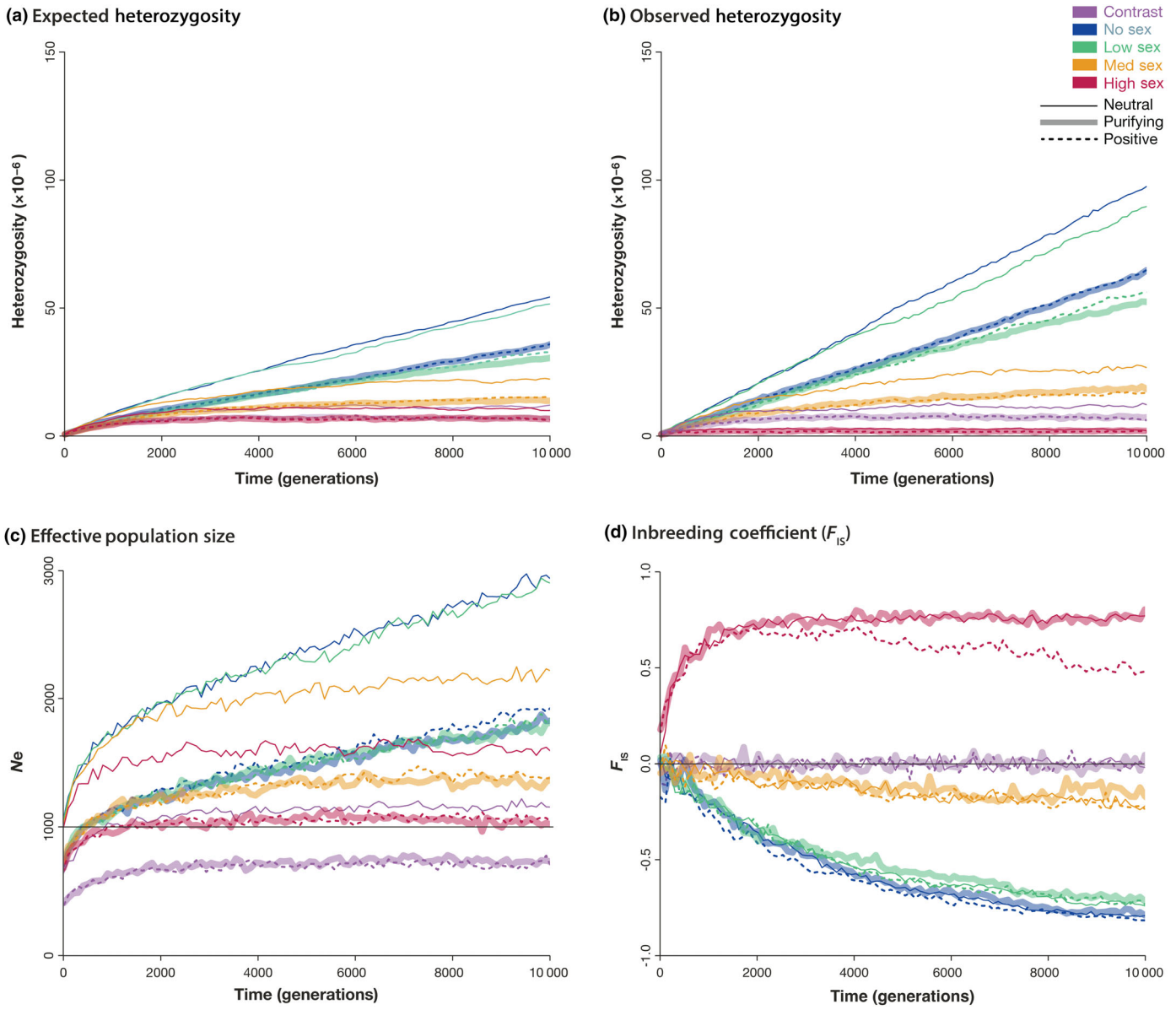


Fig. 4 Simulations with SHADIE. (a) Expected heterozygosity, (b) observed heterozygosity, (c) effective population size calculated using θ, and (d) inbreeding coefficient (F_{IS}). Line weight indicates type of selection: solid thin, neutral; dashed, positive selection; wide pale, purifying selection. Colors indicate degree of sexual reproduction (per individual per generation): red, high (1E-4); yellow, medium (1E-5), green, low (1E-6), blue, no sexual reproduction; purple, homosporous contrast. Our new *Vittaria* model was used for all simulations except for contrast simulations, which used the default homosporous pteridophyte model from SHADIE to represent typical values for ferns undergoing regular alternation of generations with sexual reproduction under the same simulation parameters. All simulations used the same chromosome model, had mutation rate of 5E-9, and population size corresponding to 2000 chromosomes (1000 diploid gametophytes individuals for *Vittaria* simulations, 2000 haploid gametophytes individuals for contrast).

heterozygosity in asexual lineages (Birky, 1996; Mark Welch & Meselson, 2000; Halkett *et al.*, 2005). We found that heterozygosity in *V. appalachiana* was greater than expected (i.e. negative F_{IS} values compared to an average F_{IS} of 0.177 across ferns) with both observed ($H_o = 0.189$) and expected ($H_S = 0.153$) heterozygosity values very similar to the average expected heterozygosity for ferns ($H_e = 0.193$; Pelosi & Sessa, 2021). The average F_{IS} of −0.226 estimated from the empirical data is similar to levels observed in asexual oribatid mites ($F_{IS} = -0.24$ to -0.39; Brandt *et al.*, 2021) and asexual kelp ($F_{IS} = -0.47$ to -0.02; Reynes *et al.*, 2021). Our simulations with SHADIE also find support for increased heterozygosity under asexual reproduction (Fig. 4b). Using the homosporous pteridophyte model with regular alternation of generations and sexual reproduction, H_o was approximately equal to H_e, and F_{IS} remained very close to 0. However, using a life cycle model realistic for *V. appalachiana*, we observed consistent $H_o > H_e$ values and strongly negative F_{IS} for simulations with very low or no sexual reproduction (Fig. 4). By contrast, simulations with moderate frequencies of sexual reproduction had F_{IS} values closer to 0, but still negative (Fig. 4).

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As predicted by Balloux *et al.* (2003), the presence of even moderate amounts of sexual reproduction in our simulations removed this pattern of $H_o > H_e$, although values of F_{IS} remained negative with all parameter combinations except for high levels of sex where F_{IS} was strongly positive (Fig. 4d). Simulations with a high frequency of sexual reproduction had highly positive F_{IS} values, with the equilibrium value of 0.73 reached by 2000 generations and maintained thereafter (Fig. 4d). To crudely parse pre- and postasexuality patterns of genetic diversity, we can subtract the parameter estimates from shadie from the empirical data, revealing that nearly all heterozygosity is very likely retained from before the loss of sexual reproduction. While the retention of heterozygosity from the ancestral sexual species may be high, our observations are consistent with the hypothesis of long-term asexual reproduction.

Expectation 3: asexual lineages should have low population differentiation As mutations accumulate in individuals and are not spread through gene flow, most variation is expected to occur within individuals, and therefore, population-level differences will account for relatively less variation. In line with this expectation, population differentiation in *V. appalachiana* was much lower ($F_{ST} = 0.017$) than the average for all ferns (average $F_{ST} = 0.281$), and substantially lower than selfing ferns (average $F_{ST} = 0.419$) and outcrossing ferns (average $F_{ST} = 0.118$; Pelosi & Sessa, 2021). Interestingly, this pattern does not appear to be universal, with high genetic differentiation between populations of the asexual kelp *Laminaria rodriguezii* ($F_{ST} = 0.18$–0.28; Reynes *et al.*, 2021) but lower differentiation in other systems, such as thrips (García-Cuenca *et al.*, 2016). Different modes of origin and age of these asexual lineages may play an important role in driving observed differences in patterns of population differentiation.

Expectation 4: asexuality should lead to increased effective population sizes Using coalescent-based modeling with δaδi, we found that the best-fitting model identified an expansion in effective population size, and subsequent decrease more recently. A similar pattern was observed in the stairwayplot analysis, where there was a rapid increase in N_e sometime during the Last Glacial Maximum, which is consistent with when the sporophyte and sexual reproduction were likely lost in this lineage. Using simulations from SHADIE, N_e was also consistently lower in the homosporous fern simulations compared with simulations using the *Vittaria* model with low rates or no sexual reproduction (Table 2). Interestingly, a gradual decrease in N_e was also observed in the stairwayplot analysis, which may be attributable to the local extinction of colonies over evolutionary time. Theory predicts that rates of clonal reproduction should be positively correlated with allelic effective population size (Balloux *et al.*, 2003), although few studies have examined this pattern in natural populations.

Using a combination of empirical data and life cycle simulations, we show that populations of *V. appalachiana* have (1) heightened levels of genomic diversity, (2) excess heterozygosity, (3) low population differentiation, and (4) increased effective population size, consistent with the theoretical consequences of prolonged clonal reproduction. Our results confirm that *V. appalachiana* experiences very little, if any, sexual reproduction, consistent with observations in the field and laboratory of its inability to produce viable sporophytes. The capacity to persist over evolutionary time in the absence of sexual reproduction likely relies on the retention and generation of genomic diversity upon which selection can act. While most other studies have focused on the sporophyte life stage of angiosperms, our study is the first to use population genomics and life cycle simulations to focus on an organism solely in the gametophytic life stage of a vascular plant. Our work demonstrates that the processes facilitating heightened genetic diversity can also occur in the haploid life phase and that the expected population genetic consequences of long-term asexual reproduction can be observed in such species. This work highlights the importance of considering the unique biology of gametophytes relative to their sporophyte counterparts, and the innovative measures through which clonal gametophytes can thrive in the absence of sex. Long-term asexual lineages demonstrate that asexuality is not an evolutionary dead end and suggest that the evolutionary disadvantages of life without sex should be reconsidered.

Demographic history and mode of origin of the Appalachian gametophyte

Determining the timing and mode of origin of *V. appalachiana* has captivated and perplexed botanists for over 50 yr (Wagner & Sharp, 1963; Farrar, 1967, 1978; Pinson & Schuettpelz, 2016). The prevailing hypothesis regarding the origin of its gametophyte-only habit has been the extinction of the sporophyte generation due to ecophysiological limitations during glaciation event(s) (e.g. intolerance to freezing or desiccation; Farrar, 1990; Pinson *et al.*, 2017). Gametophytes were traditionally assumed to be the ephemeral, ecophysiologically fragile handicap in the fern life cycle (Klekowski & Baker, 1966; Page, 2002), but a better understanding of gametophyte hardiness and tolerance in recent years has largely reversed this viewpoint. Several studies have shown that gametophytes can withstand environmental conditions inhospitable to their sporophyte counterparts (Sato & Sakai, 1981; Watkins *et al.*, 2007; Farrar *et al.*, 2008). In the case of *V. appalachiana*, gametophytes are assumed to have coped better with physiological stressors accompanying glaciation than their sporophyte counterparts (Farrar, 1978) and were also protected from extreme temperature and humidity fluctuations by their obligate habitat, sandstone rockhouse shelters (Farrar, 1998). During which glaciation period this loss of the sporophyte occurred has been unknown, with suggestions ranging from Miocene glaciation events *c.* 2 Ma to the Last Glacial Maximum 20 ka. Stevens & Emery (2015) performed a transplant experiment which showed that *V. appalachiana* gametophytes could survive in regions north of the extent of the Last Glacial Maximum, though it has not dispersed to these suitable habitats. Our stairway plot analysis (Fig. 3a) supports this timing, with an increase in effective population size *c.* 26–17 ka that is consistent with the transition from sexual to asexual reproduction. Mutations would have

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accumulated, leading to increases in effective population size in *V. appalachiana* relative to sexual taxa undergoing recombination, which was also seen in our simulations of N_e with SHADIE (Fig. 4c). Interestingly, the 1D demographic models we tested with δaδi identified older demographic events than the stairway plot, although overall the three epoch model with was the best fit (Fig. 3b). Importantly, there is an implicit assumption of some level of recombination in δaδi, which cannot be changed (but see the two-locus method described in Ragsdale & Gutenkunst, 2017); alleles may therefore appear to coalesce in fewer generations than in reality in the absence of recombination (Bengtsson, 2003).

Transitions to asexuality are commonly associated with large-scale genomic changes, such as hybridization (Bierzychudek, 1985) and/or polyploidy (Otto & Whitton, 2000). *Vittaria appalachiana* appears to follow this trend, likely originating through polyploid speciation (Gastony, 1977; Pinson & Schuettpelz, 2016; Pelosi *et al.*, 2023a). The mode of polyploidization is contentious, with earlier work with isozymes identifying fixed heterozygosity, which was interpreted as evidence for an allopolyploid origin (Farrar, 1990). However, Pinson & Schuettpelz (2016) recovered that alleles of *V. appalachiana* fell into a single clade rather than falling into separate clades with each putative diploid progenitor. The SFS generated from our empirical data lacked the distinctive peak at sites with allele frequencies of 50% (Fig. 3b), which is a feature of the SFS of segmental and full allopolyploids (Blischak *et al.*, 2023). Our findings with the SFS support the autopolyploid origin as inferred by Pinson & Schuettpelz (2016), although further cytological evidence in the form of chromosome counts, ploidy estimates, and chromosome pairing behavior is needed to fully resolve the role of polyploidy in the origin of the Appalachian gametophyte.

After transitioning to entirely asexual reproduction, existing colonies of *V. appalachiana* would presumably have become rapidly isolated, as asexual propagules (gemmae) are too heavy to be dispersed by wind. We recovered two major genetic clusters, with colonies in the southwestern-most extent of the range comprising one cluster, while central and northern colonies were mostly assigned to another (Fig. 2). A third, much smaller genetic cluster was split between Ramsey Cascades (central/south) and Rock City (north; Fig. 2). These patterns may reflect the distribution of genetic variation before the loss of sexual reproduction, or an ancient dispersal event between the central and northern populations. Despite the expectation of low dispersibility for gemmae, anecdotal evidence suggests that they may be movable other inhabitants of rockhouses, such as salamanders, toads, snails, slugs, arachnids, and insects (Farrar, 1985). In addition, Farrar (1990) showed that gemmae may disperse by water; within watersheds, downstream populations contained more genotypes, possibly due to accumulating dispersed gemmae from upstream populations. Long-distance dispersal, however, appears to be highly restricted. Further elucidation of the current patterns of genetic diversity in *V. appalachiana* would benefit from knowing (1) whether there was a single or multiple losses of sexual reproduction and (2) the level of standing genomic variation in the founding population(s).

Within colonies, expectations for patterns of genetic diversity are unclear, as the spatial pattern of gemma germination and microdistribution is unknown. A thallus producing many gemmae might drop them in the immediate vicinity of the parent, creating large patches of genetically identical individuals. However, healthy colonies leave little substrate for gemmae to establish and germinate; instead, germination may be restricted to the edges of colonies or areas where established gametophytes eventually die and free up space. If gemmae can move short distances within the colony (e.g. carried by water or animals), this could lead to a genetically heterogeneous colony, with differences even between neighboring individuals. Under the latter scenario, the sampling scheme we employed (i.e. samples consisting of multiple individual plants; Fig. 1) could lead to an artificial increase in heterozygosity compared with treating individual plants as samples. Genetically heterogeneous colonies would exacerbate this sampling effect because patches would be more likely to contain genetically distinct individuals rather than clonal relatives. Extracting DNA from individual plants was not possible for our sequencing method, although alternative techniques using tissue-direct PCR (Li *et al.*, 2010; Wu *et al.*, 2022) may be a viable approach to assessing individual-level genetic diversity in future studies.

Persistence of *V. appalachiana* and directions for future study

We have demonstrated that gametophyte-only *V. appalachiana* adheres to the major predictions for theoretical consequences of prolonged clonal reproduction and that the timing and mode of its origin and loss of sexuality are consistent with autopolyploid speciation followed by glaciation-driven loss of the sporophyte generation 26–17 ka. Despite these insights, much is still unknown about the biology of this fern, its asexual habit, and most intriguingly, how it has persisted in this state, although we highlight that the heightened genomic diversity is likely an important factor. Cryptic sex, noncanonical meiosis, and nonmeiotic homologous recombination are all possible, yet unexplored, mechanisms facilitating the long-term survival of *V. appalachiana* through the purging of deleterious alleles. The unique ploidy of *V. appalachiana*, with its homoeologous chromosomes, lends itself to preserving genetic variation and masking deleterious mutations. Cryptic sex – rare instances of sexual reproduction in presumably asexual taxa – has been proposed as a possible route preventing extinction in several ancient asexuals by purging accumulated deleterious mutations, often based on identifying genomic exchanges consistent with meiotic recombination (Gibbs & Denton, 2016; Laine *et al.*, 2022; Freitas *et al.*, 2023) or the presence of meiotic genes (Schurko & Logsdon, 2008; Chi *et al.*, 2014; but see Maciver, 2019 for a critique on this approach). Alternatively, noncanonical meiosis (Bloomfield, 2016) or nonmeiotic homologous recombination events may instead play critical roles to purge deleterious alleles to facilitate the persistence of long-term asexual lineages without sex (Simion *et al.*, 2021).

Although both functional archegonia and antheridia (with swimming sperm) have been observed on *V. appalachiana* thalli

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and germinating gemmae (Farrar, 1978; Farrar & Mickel, 1991), only a few instances of abortive sporophytes have been recorded, once in the field (Farrar, 1978) and apogamously in laboratory conditions (Farrar, 1978; Caponetti *et al.*, 1982). The production of sporophytes is therefore possible, yet seems exceedingly rare, and all observed sporophytes were developmentally stunted and never reached sufficient maturity to bear spores. Recently, we found that there was less effective selective pressure on genes related to male gamete generation in the transcriptome of *V. appalachiana* compared to closely related *Vittaria lineata* (Pelosi *et al.*, 2023a), but the molecular mechanism(s) prohibiting the production of successful sporophytes remain unknown. The loss of the sporophyte may be caused by environmental and/or genetic factors or some sequential combination of the two; for example, initial sporophyte intolerance to environmental conditions may lead to relaxed selection on genes responsible for sporophyte development, and strong selection favoring those gametophytes not using resources to produce inviable sporophytes (Farrar, 1990) causing the genetic inability to produce the diploid life stage. Additional experimentation and observational studies, such as examining chromosome pairing patterns and inducing sporophyte production, will further illuminate the mechanisms of prolonged asexual reproduction in *V. appalachiana*, and the molecular basis for the alternation of generations in general.

Conclusion

Strict asexual reproduction is relatively rare in the plant kingdom, and nearly all of what we know about this life history comes from the sporophytes of flowering plants. Our study is the first to utilize fine-scale, genome-wide molecular markers and life cycle simulations to assess how plants can persist over evolutionary time in the gametophyte life stage through obligate clonal reproduction. We show that gametophytes, once thought to be the weak link in the fern life cycle, are able to persist without a sporophyte and exhibit similar population genetic patterns to those expected under long-term asexual reproduction. Theory predicts that asexual lineages should experience declines in genetic diversity in the short term; however, in contrast with traditional thought which posits that asexuals are evolutionary dead ends, the generation of genomic diversity in the absence of sex and meiosis plays an important role in the persistence of long-term clonal plant populations. Genetic diversity is generated through the accumulation of mutations on nonrecombining homologs and through polyploidization events, facilitating the continued evolution of these asexual species.

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Competing interests

None declared.

Author contributions

JAP, ESS and EBS designed the study. JAP and ESS analyzed the data. WBB and EBS provided reagents and computing resources. JAP and EBS secured funding for the project. JAP and ESS wrote the first draft. All authors contributed to editing and revising the manuscript.

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Data availability

Demultiplexed reads have been deposited under project PRJNA1041843 at the NCBI SRA. Code and analyses are available at https://github.com/jessiepelosi/vittaria_ddRAD. Vouchered samples are available at the William and Lynda Steere Herbarium (NY) and the National Park Service (accession no. GRSM-02580), except for some samples where there was insufficient material for both DNA extraction and voucher specimens; photographs of these samples are available at the GitHub repository.

References

Aktas C. 2020. *haplotypes: manipulating DNA sequences and estimating unambiguous haplotype network with statistical parsimony.* CRAN.

Balloux F, Lehmann L, de Meeûs T. 2003. The population genetics of clonal and partially clonal diploids. *Genetics* **164**: 1635–1644.

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Barker MS. 2009. Evolutionary genomic analyses of ferns reveal that high chromosome numbers are a product of high retention and fewer rounds of polyploidy relative to angiosperms. *American Fern Journal* **99**: 136–141.

Bast J, Parker DJ, Dumas Z, Jalvingh KM, Van PT, Jaron KS, Figuet E, Brandt A, Galtier N, Schwander T. 2018. Consequences of asexuality in natural populations: insights from stick insects. *Molecular Biology and Evolution* **35**: 1668–1677.

Bengtsson BO. 2003. Genetic variation in organisms with sexual and asexual reproduction. *Journal of Evolutionary Biology* **16**: 189–199.

Bierzychudek P. 1985. Patterns in plant parthenogenesis. *Experientia* **41**: 1255–1264.

Birky CW. 1996. Heterozygosity, heteromorphy, and phylogenetic trees in asexual eukaryotes. *Genetics* **144**: 427–437.

Blischak PD, Sajan M, Barker MS, Gutenkunst RN. 2023. Demographic history inference and the polyploid continuum. *Genetics* **224**: iyad107.

Bloomfield G. 2016. Atypical ploidy cycles, Spo11, and the evolution of meiosis. *Seminars in Cell & Developmental Biology* **54**: 158–164.

Brandt A, Schaefer I, Glanz J, Schwander T, Maraun M, Scheu S, Bast J. 2017. Effective purifying selection in ancient asexual oribatid mites. *Nature Communications* **8**: 873.

Brandt A, Tran Van P, Bluhm C, Anselmetti Y, Dumas Z, Figuet E, François CM, Galtier N, Heimburger B, Jaron KS *et al.* 2021. Haplotype divergence supports long-term asexuality in the oribatid mite *Oppiella nova*. *Proceedings of the National Academy of Sciences, USA* **118**: e2101485118.

Caponetti JD, Whitten M, Beck MJ. 1982. Axenic culture and induction of callus and sporophytes of the Appalachian *Vittaria* gametophyte. *American Fern Journal* **72**: 36–40.

Chambers SM, Emery NC. 2016. Population differentiation and countergradient variation throughout the geographic range in the fern gametophyte *Vittaria appalachiana*. *American Journal of Botany* **103**: 86–98.

Chambers SM, Watkins JE, Sessa EB. 2017. Differences in dehydration tolerance among populations of a gametophyte-only fern. *American Journal of Botany* **104**: 598–607.

Chapman RH, Klekowski EJ, Selander RK. 1979. Homoeologous heterozygosity and recombination in the fern *Pteridium aquilinum*. *Science* **204**: 1207–1209.

Chen CW, Huang YM, Kuo LY, Chang YH, Liu YC, Chiou WL. 2013a. A new vittarioid fern species, *Haplopteris heterophylla* (Pteridaceae). *Systematic Botany* **38**: 901–909.

Chen CW, Huang YM, Kuo LY, Nguyen QD, Luu HT, Callado JR, Farrar DR, Chiou WL. 2013b. trnL-F is a powerful marker for DNA identification of field vittarioid gametophytes (Pteridaceae). *Annals of Botany* **111**: 663–673.

Chi J, Parrow MW, Dunthorn M. 2014. Cryptic sex in *Symbiodinium* (Alveolata, Dinoflagellata) is supported by an inventory of meiotic genes. *Journal of Eukaryotic Microbiology* **61**: 322–327.

Dalton AS, Margold M, Stokes CR, Tarasov L, Dyke AS, Adams RS, Allard S, Arends HE, Atkinson N, Attig JW *et al.* 2020. An updated radiocarbon-based ice margin chronology for the last deglaciation of the North American Ice Sheet Complex. *Quaternary Science Reviews* **234**: 106223.

Danecek P, Auton A, Abecasis G, Albers CA, Banks E, DePristo MA, Handsaker RE, Lunter G, Marth GT, Sherry ST *et al.* 2011. The variant call format and VCFTOOLS. *Bioinformatics* **27**: 2156–2158.

Danecek P, Bonfield JK, Liddle J, Marshall J, Ohan V, Pollard MO, Whitwham A, Keane T, McCarthy SA, Davies RM *et al.* 2021. Twleve years of SAMTOOLS and BCFTOOLS. *GigaScience* **10**: giab008.

Eaton DAR, Overcast I. 2020. IPYRAD: interactive assembly and analysis of RADseq datasets. *Bioinformatics* **36**: 2592–2594.

Ebihara A, Yamaoka A, Mizukami N, Sakoda A, Nitta JH, Imaichi R. 2013. A survey of the fern gametophyte flora of Japan: frequent independent occurrences of noncordiform gametophytes. *American Journal of Botany* **100**: 735–743.

Ellstrand NC, Roose ML. 1987. Patterns of genotypic diversity in clonal plant species. *American Journal of Botany* **74**: 123–131.

Excoffier L, Smouse PE, Quattro JM. 1992. Analysis of molecular variance inferred from metric distances among DNA haplotypes: application to human mitochondrial DNA restriction data. *Genetics* **131**: 479–491.

Farrar DR. 1967. Gametophytes of four tropical fern genera reproducing independently of their sporophytes in the southern Appalachians. *Science* **155**: 1266–1267.

Farrar DR. 1978. Problems in the identity and origin of the Appalachian *Vittaria* gametophyte, a sporophyteless fern of the eastern United States. *American Journal of Botany* **65**: 1–12.

Farrar DR. 1985. Independent fern gametophytes in the wild. *Proceedings of the Royal Society of Edinburgh. Section B: Biological Sciences* **86**: 361–369.

Farrar DR. 1990. Species and evolution in asexually reproducing independent fern gametophytes. *Systematic Botany* **15**: 98–111.

Farrar DR. 1998. The tropical flora of rockhouse cliff formations in the Eastern United States. *Journal of the Torrey Botanical Society* **125**: 91–108.

Farrar DR, Dassler CL, Watkins JE Jr, Skelton C. 2008. Gametophyte ecology. In: Ranker TA, Haufler CH, eds. *Biology and evolution of ferns and lycophytes*. London, UK: Cambridge University Press, 222–256.

Farrar DR, Mickel JT. 1991. *Vittaria appalachiana*: a name for the "Appalachian gametophyte". *American Fern Journal* **81**: 69–75.

Foulley J-L, Ollivier L. 2006. Estimating allelic richness and its diversity. *Livestock Science* **101**: 150–158.

Freitas S, Parker DJ, Labédan M, Dumas Z, Schwander T. 2023. Evidence for cryptic sex in parthenogenetic stick insects of the genus *Timema*. *Proceedings of the Royal Society B* **290**: 20230404.

García-Cuenca AF, Dumas Z, Schwander T. 2016. Extreme genetic diversity in asexual grass thrips populations. *Journal of Evolutionary Biology* **29**: 887–899.

Garrison E, Marth G. 2012. Haplotype-based variant detection from short-read sequencing. *arXiv*: 1207.3907.

Gastony GJ. 1977. Chromosomes of the independently reproducing Appalachian gametophyte: a new source of taxonomic evidence. *Systematic Botany* **2**: 43–48.

Gibbs HL, Denton RD. 2016. Cryptic sex? Estimates of genome exchange in unisexual mole salamanders (*Ambystoma* sp.). *Molecular Ecology* **25**: 2805–2815.

Goudet J. 2005. HIERFSTAT, a package for R to compute and test hierarchical F-statistics. *Molecular Ecology Notes* **5**: 184–186.

Grusz AL, Rothfels CJ, Schuettpelz E. 2016. Transcriptome sequencing reveals genome-wide variation in molecular evolutionary rate among ferns. *BMC Genomics* **17**: 692.

Gutenkunst RN, Hernandez RD, Williamson SH, Bustamante CD. 2009. Inferring the joint demographic history of multiple populations from multidimensional SNP frequency data. *PLoS Genetics* **5**: e1000695.

Halkett F, Simon J-C, Balloux F. 2005. Tackling the population genetics of clonal and partially clonal organisms. *Trends in Ecology & Evolution* **20**: 194–201.

Hartl DL, Clark AG. 1989. *Principles of population genetics*. Sunderland, MA, USA: Sinauer Associates.

Haufler CH, Pryer KM, Schuettpelz E, Sessa EB, Farrar DR, Moran R, Schneller JJ, Watkins JE, Windham MD. 2016. Sex and the single gametophyte: revising the homosporous vascular plant life cycle in light of contemporary research. *Bioscience* **66**: 928–937.

Ho KHE, Bartkowska M, Wright SI, Agrawal AF. 2019. Population genomics of the facultatively asexual duckweed *Spirodela polyrhiza*. *New Phytologist* **224**: 1361–1371.

Holsinger KE. 2000. Reproductive systems and evolution in vascular plants. *Proceedings of the National Academy of Sciences, USA* **97**: 7037–7042.

Jaron KS, Bast J, Nowell RW, Ranallo-Benavidez TR, Robinson-Rechavi M, Schwander T. 2021. Genomic features of parthenogenetic animals. *Journal of Heredity* **112**: 19–33.

Johnson SG. 2006. Geographic ranges, population structure, and ages of sexual and parthenogenetic snail lineages. *Evolution* **60**: 1417–1426.

Jombart T. 2008. ADEGENET: a R package for the multivariate analysis of genetic markers. *Bioinformatics* **24**: 1403–1405.

Judson OP, Normark BB. 1996. Ancient asexual scandals. *Trends in Ecology & Evolution* **11**: 41–46.

Kamvar ZN, Brooks JC, Grünwald NJ. 2015. Novel R tools for analysis of genome-wide population genetic data with emphasis on clonality. *Frontiers in Genetics* **6**: 208.

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Kinosian SP, Testo WL, Chambers SM, Sessa EB. 2019. Using RAD data to confirm parentage of polyploids in a reticulate complex of ferns. *American Fern Journal* **109**: 267–282.

Klekowski EJ, Baker HG. 1966. Evolutionary significance of polyploidy in the Pteridophyta. *Science* **153**: 305–307.

Koffi M, De Meeûs T, Bucheton B, Solano P, Camara M, Kaba D, Cuny G, Ayala FJ, Jamonneau V. 2009. Population genetics of *Trypanosoma brucei gambiense*, the agent of sleeping sickness in Western Africa. *Proceedings of the National Academy of Sciences, USA* **106**: 209–214.

Korunes KL, Samuk K. 2021. Pixy: unbiased estimation of nucleotide diversity and divergence in the presence of missing data. *Molecular Ecology Resources* **21**: 1359–1368.

Kuo L-Y, Chen C-W, Shinohara W, Ebihara A, Kudoh H, Sato H, Huang Y-M, Chiou W-L. 2017. Not only in the temperate zone: independent gametophytes of two vittarioid ferns (Pteridaceae, Polypodiales) in East Asian subtropics. *Journal of Plant Research* **130**: 255–262.

Laczkó L, Jordán S, Sramkó G. 2022. The RadOrgMiner pipeline: automated genotyping of organellar loci from RADseq data. *Methods in Ecology and Evolution* **13**: 1962–1975.

Laine VN, Sackton TB, Meselson M. 2022. Genomic signature of sexual reproduction in the bdelloid rotifer *Macrotrachella quadricornifera*. *Genetics* **220**: iyab221.

Li FW, Kuo LY, Huang YM, Chiou WL, Wang CN. 2010. Tissue-direct PCR, a rapid and extraction-free method for barcoding of ferns. *Molecular Ecology Resources* **10**: 92–95.

Li H, Durbin R. 2009. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* **25**: 1754–1760.

Liang S-Q, Zhang Z-C, Wei R. 2022. Ecological adaptation shaped the genetic structure of homoploid ferns against strong dispersal capacity. *Molecular Ecology* **31**: 2679–2697.

Liu X, Fu Y-X. 2020. Stairway Plot 2: demographic history inference with folded SNP frequency spectra. *Genome Biology* **21**: 280.

Maciver SK. 2019. Ancestral eukaryotes reproduced asexually, facilitated by polyploidy: a hypothesis. *BioEssays: News and Reviews in Molecular, Cellular and Developmental Biology* **41**: e1900152.

Marchant DB, Chen G, Cai S, Chen F, Schafran P, Jenkins J, Shu S, Plott C, Webber J, Lovell JT *et al.* 2022. Dynamic genome evolution in a model fern. *Nature Plants* **8**: 1038–1051.

Mark Welch D, Meselson M. 2000. Evidence for the evolution of bdelloid rotifers without sexual reproduction or genetic exchange. *Science* **288**: 1211–1215.

Mark Welch DB, Meselson MS. 2001. Rates of nucleotide substitution in sexual and anciently asexual rotifers. *Proceedings of the National Academy of Sciences, USA* **98**: 6720–6724.

Maynard Smith J. 1978. *The evolution of sex*. London, UK: Cambridge University Press.

McKenna A, Hanna M, Banks E, Sivachenko A, Cibulskis K, Kernytsky A, Garimella K, Altshuler D, Gabriel S, Daly M *et al.* 2010. The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. *Genome Research* **20**: 1297–1303.

Neiman M, Meirmans S, Meirmans PG. 2009. What can asexual lineage age tell us about the maintenance of sex? *Annals of the New York Academy of Sciences* **1168**: 185–200.

Nelder JA, Mead R. 1965. A simple method for function minimization. *The Computer Journal* **7**: 308–313.

Nitta JH, Schuettpelz E, Ramírez-Barahona S, Iwasaki W. 2022. An open and continuously updated fern tree of life. *Frontiers in Plant Science* **13**: 909768.

Otto SP, Whitton J. 2000. Polyploid incidence and evolution. *Annual Review of Genetics* **34**: 401–437.

Page CN. 2002. Ecological strategies in fern evolution: a neopteridological overview. *Review of Palaeobotany and Palynology* **119**: 1–33.

Paradis E. 2010. pegas: an R package for population genetics with an integrated-modular approach. *Bioinformatics* **26**: 419–420.

Pelosi JA, Barbazuk WB, Sessa EB. 2023a. Life without a sporophyte: the origin and genomic consequences of asexual reproduction in a gametophyte-only fern. *International Journal of Plant Sciences* **184**: 454–469.

Pelosi JA, Sessa EB. 2021. From genomes to populations: a meta-analysis and review of fern population genetics. *International Journal of Plant Sciences* **182**: 325–343.

Pelosi JA, Zumwalde BA, Sorojsrisom ES, Sessa EB. 2023b. A genome size for the Appalachian gametophyte. *American Fern Journal* **113**: 257–262.

Pelosi JA, Zumwalde BA, Testo WL, Kim EH, Burleigh JG, Sessa EB. 2024. All tangled up: unraveling phylogenetics and reticulate evolution in the vining ferns, *Lygodium* (Schizaeales). *American Journal of Botany* **111**: e16389.

Pinson JB, Chambers SM, Nitta JH, Kuo L-Y, Sessa EB. 2017. The separation of generations: biology and biogeography of long-lived sporophyteless fern gametophytes. *International Journal of Plant Sciences* **178**: 1–18.

Pinson JB, Schuettpelz E. 2016. Unraveling the origin of the Appalachian gametophyte, *Vittaria appalachiana*. *American Journal of Botany* **103**: 668–676.

Portik DM, Leaché AD, Rivera D, Barej MF, Burger M, Hirschfeld M, Rödel M-O, Blackburn DC, Fujita MK. 2017. Evaluating mechanisms of diversification in a Guineo-Congolian tropical forest frog using demographic model selection. *Molecular Ecology* **26**: 5245–5263.

Qiu Y-L, Taylor AB, McManus HA. 2012. Evolution of the life cycle in land plants. *Journal of Systematics and Evolution* **50**: 171–194.

R Core Team. 2022. *R: a language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/

Ragsdale AP, Gutenkunst RN. 2017. Inferring demographic history using two-locus statistics. *Genetics* **206**: 1037–1048.

Reynes L, Thibaut T, Mauger S, Blanfuné A, Holon F, Cruaud C, Couloux A, Valero M, Aurelle D. 2021. Genomic signatures of clonality in the deep water kelp *Laminaria rodriguezii*. *Molecular Ecology* **30**: 1806–1822.

Ricci CN. 1987. Ecology of bdelloids: how to be successful. *Hydrobiologia* **147**: 117–127.

Rice P, Longden I, Bleasby A. 2000. Emboss: the European Molecular Biology Open Software Suite. *Trends in Genetics* **16**: 276–277.

Robison TA, Grusz AL, Wolf PG, Mower JP, Fauskee BD, Sosa K, Schuettpelz E. 2018. Mobile elements shape plastome evolution in ferns. *Genome Biology and Evolution* **10**: 2558–2571.

Rothfels CJ, Schuettpelz E. 2014. Accelerated rate of molecular evolution for vittarioid ferns is strong and not driven by selection. *Systematic Biology* **63**: 31–54.

Sato T, Sakai A. 1981. Cold tolerance of gametophytes and sporophytes of some cool temperate ferns native to Hokkaido. *Canadian Journal of Botany* **59**: 604–608.

Schurko AM, Logsdon JM. 2008. Using a meiosis detection toolkit to investigate ancient asexual "scandals" and the evolution of sex. *BioEssays: News and Reviews in Molecular, Cellular and Developmental Biology* **30**: 579–589.

Schwander T, Crespi BJ. 2009. Twigs on the tree of life? Neutral and selective models for integrating macroevolutionary patterns with microevolutionary processes in the analysis of asexuality. *Molecular Ecology* **18**: 28–42.

Schwander T, Crespi BJ, Gries R, Gries G. 2013. Neutral and selection-driven decay of sexual traits in asexual stick insects. *Proceedings of the Royal Society B* **280**: 20130823.

Simion P, Narayan J, Houtain A, Derzelle A, Baudry L, Nicolas E, Arora R, Cariou M, Cruaud C, Gaudray FR *et al.* 2021. Chromosome-level genome assembly reveals homologous chromosomes and recombination in asexual rotifer *Adineta vaga*. *Science Advances* **7**: eabg4216.

Simon JC, Delmotte F, Rispe C, Crease T. 2003. Phylogenetic relationships between parthenogens and their sexual relatives: the possible routes to parthenogenesis in animals. *Biological Journal of the Linnean Society* **79**: 151–163.

Sorojsrisom ES, Haller BC, Ambrose BA, Eaton DAR. 2022. Selection on the gametophyte: modeling alternation of generations in plants. *Applications in Plant Sciences* **10**: e11472.

Stevens SM, Emery NC. 2015. Dispersal limitation and population differentiation in performance beyond a northern range limit in an asexually reproducing fern. *Diversity and Distributions* **21**: 1242–1253.

Testo W, Sundue M. 2016. A 4000-species dataset provides new insight into the evolution of ferns. *Molecular Phylogenetics and Evolution* **105**: 200–211.

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Wagner WH, Sharp AJ. 1963. A remarkably reduced vascular plant in the United States. *Science* **142**: 1483–1484.
Watkins JE, Mack M, Sinclair TR, Mulkey SS. 2007. Ecological and evolutionary consequences of desiccation tolerance in tropical fern gametophytes. *New Phytologist* **176**: 708–717.
Winter DJ. 2012. MMOD: an R library for the calculation of population differentiation statistics. *Molecular Ecology Resources* **12**: 1158–1160.
de Witte LC, Stöcklin J. 2010. Longevity of clonal plants: why it matters and how to measure it. *Annals of Botany* **106**: 859–870.
Wu Y-H, Ke Y-T, Chan Y-Y, Wang G-J, Kuo L-Y. 2022. Integrating tissue-direct PCR into genetic identification: an upgraded molecular ecology approach to survey fern gametophytes in the field. *Applications in Plant Sciences* **10**: e11462.

Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Dataset S1 Voucher information, nuclear and plastid assembly statistics, and detailed results.

Dataset S2 Parameters used in shadie simulations.

Fig. S1 Intra- and inter-colony and population genetic distances and clone correction.

Fig. S2 SliM event sequence for *Vittaria* model in shadie.

Fig. S3 *Vittaria appalachiana* life cycle simulation parameters in shadie.

Fig. S4 Stairway plot estimates of changes in effective population size with altering parameters.

Fig. S5 1D demographic model results with δaδi.

Fig. S6 Plastid haplotype network.

Fig. S7 BIC values of *K* for DAPC clustering.

Notes S1 Analysis of ipyrad variants.

Notes S2 Clone correction analysis.

Notes S3 A note on homeologous heterozygosity.

Table S1 AMOVA with populations, colonies, and individuals.

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